

ZAYAAN BHANWADIA

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EDUCATION

University of Toronto

Sept. 2025 – Dec. 2028

Honours B.Sc. in Computer Science (Co-op) | GPA: 3.7/4.0

Toronto, ON

EXPERIENCE

Software Engineer Intern (*Incoming*)

Sept. 2026 – Dec 2026

FreshBooks

Toronto, ON

- Incoming intern to develop backend microservices in Python and UI components with React for 10M+ users.
- Set to deploy applications using Docker and Azure DevOps with RESTful APIs and MySQL/Postgres databases.
- Will collaborate in an agile team to automate testing, optimize data via Redis, and ship production features.

Software Engineer Intern

May 2026 – Present

KorraNet Creative

Winnipeg, Manitoba (*Remote*)

- Restructured secret ingestion and eliminated hardcoded credentials across **3 core services**, migrating to secure environment management and reducing credential exposure.
- Optimized Meta OAuth token handling and storage architecture, reducing token refresh failures by **40%+** and ensuring reliable context availability for downstream AI model integrations.
- Reduced system crashes and failed API calls by **85%+** through adding API error handling and timeout prevention.
- Implemented logging across onboarding and authentication flows to isolate bugs and support new developer teams.

Co-President & Lead Engineer

Sept. 2023 – June 2025

NASA Hunch

Toronto, Ontario

- Secured a **3rd-place international** finish as the executive of the first Canadian team to compete in NASA Hunch.
- Built an autonomous Lunar Bot using Arduino and CAD, achieving **95%+ obstacle avoidance** in simulated terrain.
- Programmed a real-time C++ control algorithm using non-blocking execution loops to synchronize timed image capture and prevent system unresponsiveness.

PROJECTS

Embedded Key-Value Store | C++, POSIX Threads, Linux

June 2026

- Built an embedded C++ database with data sharding to prevent global lock bottlenecks and improve throughput.
- Implemented shared locks for non-blocking reads across **4 threads**, preventing race conditions during writes.
- Designed data persistence by writing updates to a log file before memory to ensure crash recovery and durability.
- Created a file compaction routine to safely clean up old or deleted keys and **reduce disk usage by 45%+** without pausing the app.

Context Sync Extension | TypeScript, VS Code Extension API

May 2026

- Developed a VS Code extension with **500+ downloads** that automates AI chat context sharing to maintain workspace continuity across developer environments.
- Designed a graph-based Markdown schema to maximize context density per token across AI assistant sessions.
- Integrated GitHub Copilot, OneDrive, and Google Drive to automate syncing and enable one-click environment setup.

Clinical AI Assistant (Top 10 Projects @ GenAI Genesis) | Python, SQL, OpenCV

Mar. 2026

- Deployed an AI-powered clinical assistant to automate patient check-ins and intake, and organized data into a schema.
- Engineered a live monitoring dashboard with computer vision, achieving **98%+ accuracy** for real-time fall detection.
- Developed a spatial visualization algorithm that generates 3D labelled hospital models by extruding floor plans.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C, SQL, Bash, HTML/CSS

Libraries/Frameworks: FastAPI, Node.js, Flask, SQLAlchemy, React.js, Next.js, OpenCV, NumPy

Developer Tools: Linux, Docker, Git, Github, PGAdmin, VS Code, Jira/Confluence, Slack

Cloud/Databases: PostgreSQL, IBM Cloud, Google Cloud Project, Vercel, Vultr, Railway, Supabase